

EduGenie: Google Gemini Powered Learning Assistant

1. Executive Summary & Overview

EduGenie is an intelligent, interactive learning assistant designed to transform complex study materials into accessible, tailored educational content. Powered by Google's state-of-the-art `gemini-2.5-flash` model via the official `google-genai` SDK, EduGenie addresses the core challenges faced by modern learners: information overload, difficult technical concepts, and static study resources.

2. Step-by-Step Implementation Guide

Step 1: Install Required Libraries

Install the official Google GenAI SDK using pip:

```
pip install google-genai
```

Step 2: Set Up API Key

Obtain an API key from Google AI Studio and configure it in your environment:

- **Linux/macOS:** `export GEMINI_API_KEY="your_api_key_here"`
- **Windows:** `set GEMINI_API_KEY=your_api_key_here`

3. Python Source Code

```
import os
```

```
from google import genai
```

```
# Setup API key
```

```
api_key = os.environ.get("GEMINI_API_KEY", "YOUR_GEMINI_API_KEY")
```

```
# Initialize Client
```

```

client = genai.Client(api_key=api_key)

def edugenie_assistant(user_prompt: str) → str:

    """Function to interact with EduGenie Learning Assistant"""

    try:

        system_context = (

            "You are EduGenie, an encouraging and intelligent AI Learning Assistant. "

            "Explain concepts clearly in simple terms with examples. "

            f"User Query: {user_prompt}"

        )

        response = client.models.generate_content(

            model="gemini-2.5-flash",

            contents=system_context,

        )

        return response.text

    except Exception as e:

        return f"An error occurred: {str(e)}"

if __name__ == "__main__":

    print("--- Welcome to EduGenie AI Learning Assistant ---\n")

    student_query = "What is Photosynthesis? Explain in simple terms."

    print(f"User Query: {student_query}\n")

```

```
ai_response = edugenie_assistant(student_query)
```

```
print("--- EduGenie Response ---")
```

```
print(ai_response)
```

4. Sample Output

--- Welcome to EduGenie AI Learning Assistant ---

User Query: What is Photosynthesis? Explain in simple terms.

--- EduGenie Response ---

Photosynthesis is the process that green plants use to create their own food using energy from sunlight!

Here is how it works step-by-step:

1. Sunlight: Plants capture light energy using a green pigment in their leaves called chlorophyll.
2. Water: Roots absorb water from the soil and transport it up to the leaves.
3. Carbon Dioxide: Leaves take in carbon dioxide gas from the surrounding air through tiny openings called stomata.

The Chemical Equation:

Carbon Dioxide + Water + Sunlight → Glucose (Food) + Oxygen

Simple Analogy:

Think of a plant leaf as a tiny solar-powered kitchen! Sunlight acts as the stove, water and carbon dioxide are the raw ingredients, and glucose is the delicious meal that helps the plant grow.

5. Conclusion & Future Scope

- **Efficiency:** Powered by the `gemini-2.5-flash` model, EduGenie delivers rapid and precise explanations.
- **Educational Value:** Enhances learning by converting abstract concepts into intuitive analogies.
- **Future Enhancements:** Support for multimodal inputs (e.g., diagram analysis), interactive quizzes, and voice support.